ASSIGNMENT-01

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Task-1 (Mathematical Formulation)

1. Alphabet : $\Sigma = \{0,1\}$
2. String : A finite Sequence of symbols Ex : 101
3. Language : A set of strings over an alphabet.

   Ex : $L = \{0, 01, 101\}$

4. types :

   finite string : 1010

   Empty String : ε, length = 0

   Language : $L \subseteq \Sigma^*$

5. finite Automation :

   $M = (Q, \Sigma, \delta, q_0, F)$.

   Q – set of states

   Σ – input Alphabet

   δ – Transition function

   q_0 – initial state

   F – final states

6. Sample Strings: Language of strings ending in 1

String → Result

101 → Accepted

111 → Accepted

100 → Rejected

010 → Rejected

1 → Accepted.

TASK2: DFA - strings Starting & Ending with Same Symbol

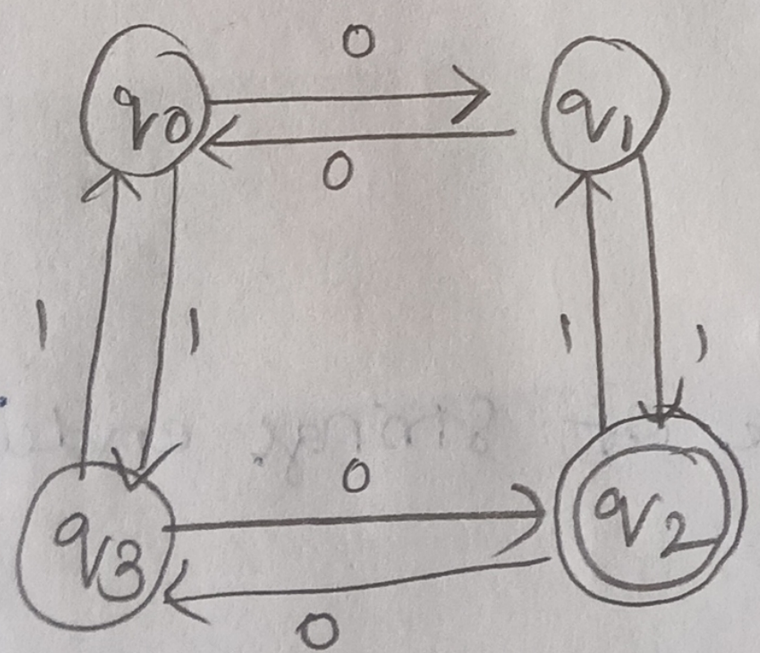
$M = (Q, \Sigma, \delta, q_s, F)$

$Q = \{q_s, q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$, $F = \{q_0, q_2\}$

Transition table.

State	0	1
→ q_s	q_0	q_2
* q_0	q_0	q_1
q_1	q_0	q_1
* q_2	q_3	q_2
q_3	q_3	q_2



Tracing

String	Result
101	Accepted
100	Rejected
010	Accepted
011	Rejected

Conclusion: The DFA accepts strings that start & end with the same symbol

TASK3 - Non-Determin
Pattern Recognition

1. Stately
• q_0 - initial
q_1 - f

TASK 3 - Non-Deterministic finite Automata for Pattern Recognition

1. States
   - q_0 - initial State
     q_1 - first 1 detected
     q_2 - Sub String 10 detected
     q_3 - substring 101 detected (final State)

2. five-tuple representation:

$$M = (Q, \Sigma, \delta, q_0, F)$$

$$Q = \{q_0, q_1, q_2, q_3\}$$

$$\Sigma = \{0, 1\} \qquad F = \{q_3\}$$

3. Transition Table:

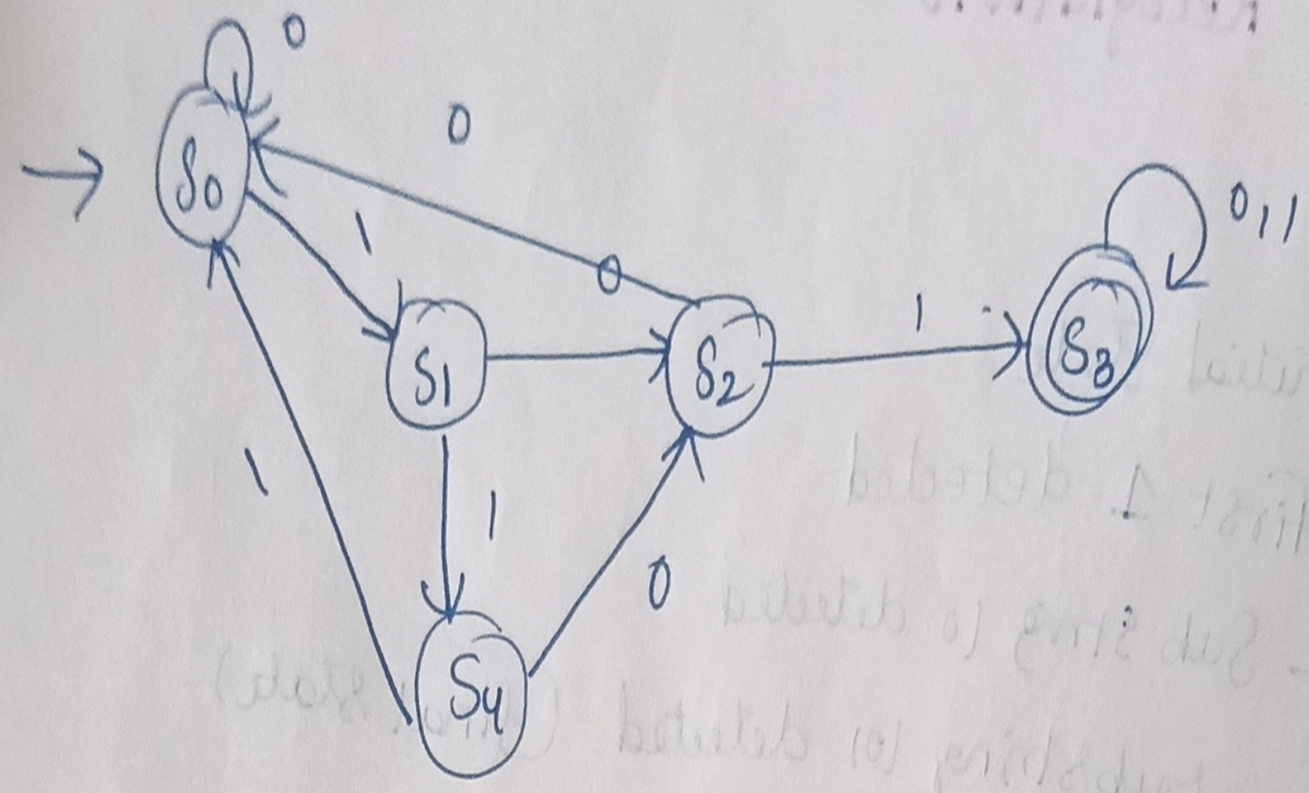
State	0	1
$\rightarrow q_0$	$\{q_0\}$	$\{q_0, q_1\}$
q_1	$\{q_2\}$	ϕ
q_2	ϕ	$\{q_3\}$
$*q_3$	$\{q_3\}$	$\{q_3\}$

100 → rejected
111 → rejected

Tracing:

String	State Sequence	Result
101	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$	Accepted
1101	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_3$	Accepted

State Transition diagram:



Task 4: Equivalence of NFA & DFA Using subset Construction

1) NFA States

$$Q = \{q_0, q_1, q_2, q_3\}$$

2) Subset Construction

let $A = \{q_0\}$, $B = \{q_0, q_1\}$

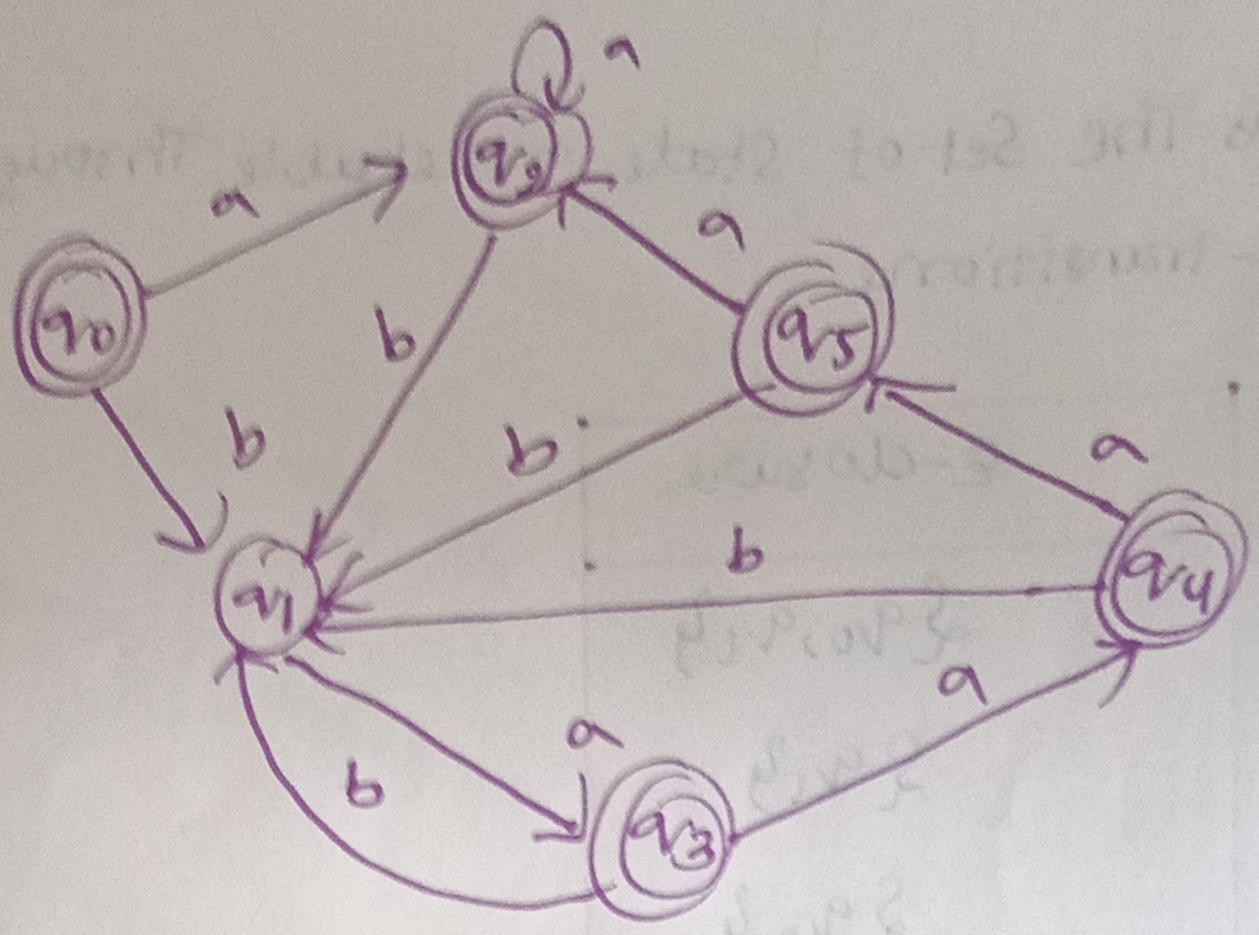
$C = \{q_0, q_2\}$, $D = \{q_0, q_1, q_3\}$.

$E = \{q_0, q_2, q_3\}$, $F = \{q_0, q_3\}$

3) DFA transition table:

DFA State	0	1
→ A	A	B
B	C	B
C	A	D
* D	E	D
* E	F	D
* F	F	D.

final states: D, E, F (subset contains q_3)



101 → Accepted } NFA & DFA
1101 → accepted

100 } Rejected in NFA & DFA
111

TASK 5 - ε NFA construction, ε-closure & Equivalence Analysis

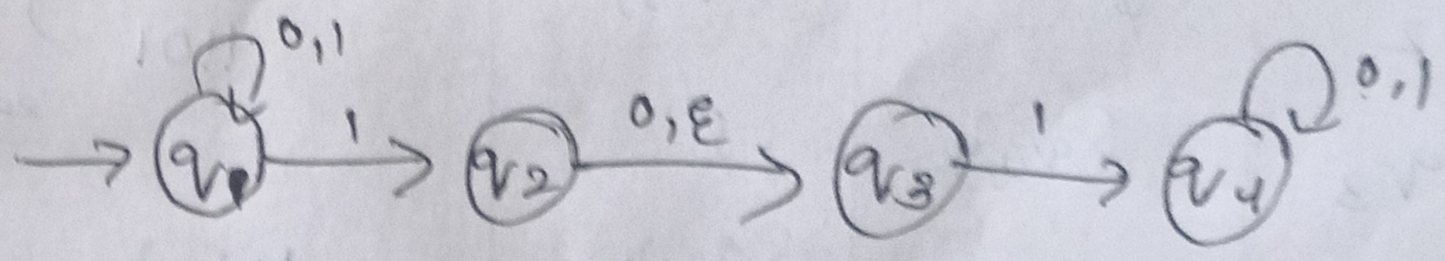
1. ε-NFA construction

States :

$Q = \{q_0, q_1, q_2, q_3, q_4\}$ ← q_4 → final state

↓ initial state

2. State Transition Diagram



This automaton accepts "0110", because there is a

possible path that leads to an accepting state

3) ε-closure.

ε-closure is the set of state reachable through

0 or more ε-transitions

State	ε-closure
q_0	$\{q_0, q_1\}$
q_1	$\{q_1\}$
q_2	$\{q_2\}$
q_3	$\{q_3\}$
q_4	$\{q_4\}$

4) Equivalent NFA without ε-Transitions.

$\delta'(q,a) = \varepsilon\text{-closure}(\delta(\varepsilon\text{-closure}(q), a))$

Transition table

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_1, q_2\}$
q_1	ϕ	$\{q_2\}$
q_2	ϕ	$\{q_3\}$
q_3	$\{q_4\}$	ϕ
$*q_4$	$\{q_4\}$	$\{q_4\}$
		$f' = \{q_4\}$

i/p

Verification

110, 0110, 1101 } Accepted ε-NFA equivalent NFA

101, 100 } rejected in both